

HUNG-YANG CHANG

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PROFESSIONAL SUMMARY

AI Software Engineer with 5+ years delivering production-grade automotive AI systems end-to-end—from LLM orchestration and microservices architecture to observability and reliability at scale. Drove $9\times$ inference throughput gains and 40% latency reduction; built telemetry pipelines tracking 1M+ monthly commands. Published researcher (IEEE, ACM) in edge computing and neural architecture search.

PROFESSIONAL EXPERIENCE

Cloud Platform Innovations team, Cerence (Hybrid) Montreal, Canada
AI Software Developer, Full-time (Permanent) Aug. 2023 – Present

- Architected and shipped the LLM function-calling orchestration pipeline for **Cerence XUI**, serving 1M+ monthly car-control commands across production automotive platforms for major OEMs
- Led end-to-end design of a gRPC-based multi-agent microservices architecture, enabling asynchronous task orchestration while sustaining 99.9% service uptime in production
- Engineered a multi-layer automated test framework (unit, contract, integration, E2E) with Pytest and ReportPortal dashboards, enforcing CI/CD quality gates and achieving 95%+ code coverage
- Established end-to-end telemetry observability for 1M+ monthly XUI Gen 2 car-control commands, enabling latency tracking and performance analysis across Azure pipelines with Databricks analytics

Bittensor (Open-source ML), Opentensor (Remote) Ontario, Canada
Machine Learning Engineer, full-time contract with Mining team Mar. 2023 - Jul. 2023

- Fine-tuned and ensembled multiple LLMs on the 🌀 **Bittensor decentralized network**, achieving top-tier text-generation scores that surpassed GPT-4 under the Bittensor reward mechanism
- Designed the multi-modal subnet architecture for the Bittensor network, extending platform capabilities across text-to-image, text-to-video, and text-to-music generation pipelines

McGill Edge Intelligence Lab, McGill Quebec, Canada
Graduate Research Assistant, advised by Professor Warren Gross, full-time Sep. 2020 - Feb. 2023

- Proposed a heterogeneous pipeline framework for edge BERT inference, achieving 49% higher throughput and 61% lower energy-delay product vs. the best homogeneous baseline 📄 **JSPS'22**
- Integrated Neural Architecture Search with pipeline parallelism on BERT, delivering $9\times$ inference throughput improvement with only 1.3% accuracy loss 📄 **EIW'22** 📄 **GLSVLSI'23**

Neuromorphic Devices and Architectures Research Group, IBM San Jose, CA, USA
Research Intern, mentored by Dr. Geoffrey W. Burr, full-time Oct. 2018 - Apr. 2019

- Optimized analog AI accelerator circuit architecture, reducing power consumption to achieve 12–14 TOPS/W energy efficiency for DNN training workloads 📄 **IBM Journal of R&D'19**

EDUCATION

McGill University, Canada Sep. 2020 - Feb. 2023
MSc (thesis) in Electrical & Computer Engineering Overall GPA: 4/4

National Tsing Hua University, Hsinchu (NTHU), Taiwan Sep. 2014 - Jun. 2018
B.S. in Electrical Engineering Overall GPA: 3.99/4.3 (3.87/4)

SELECTED PUBLICATION

PipeBERT: High-throughput BERT Inference for ARM Big.LITTLE Multi-core Processors, *Journal of Signal Processing Systems, IEEE SiPS 2022*

- Hung-Yang Chang, Seyyed Mozafari, Cheng Chen, James Clark, Brett Meyer, and Warren Gross

AI hardware acceleration with analog memory: micro-architectures for low energy at high speed, *IBM Journal of Research and Development*

- Hung-Yang Chang and Geoffrey W. Burr, Pritish Narayanan, Stefano Ambrogio, et al.

High-Throughput Edge Inference for BERT Models via Neural Architecture Search and Pipeline, GLSVLSI 2023 (Poster Presentation)

- Hung-Yang Chang, Seyyed Mozafari, James Clark, Brett Meyer, and Warren Gross

A Novel Architecture to Build Ideal-linearity Neuromorphic Synapses on a Pure Logic FinFET Platform Featuring 2.5ns PGM-time and 10^{12} Endurance, 2019 Symposium on VLSI Technology (Oral Presentation)

- E.R Hsieh, H. Y. Chang, Steve S. Chung, S. Simon Wong et al.

SELECTED PROJECTS — FULL PORTFOLIO

- Puppy Step — Companion app tracking pet growth and milestones with real-time updates
- Subscribe Manager — Recurring expense tracker with intelligent subscription insights
- Canada Count Down — Timeline tracker from Permanent Residency to Canadian Citizenship
- FoundIt — Lost & found tracker using QR codes and real-time messaging
- Coach Book IQ — Smart team management and scheduling system for coaching staff

AWARDS & HONORS

Graduate Excellence Fellowship

ECSE, McGill, 2022

- Awarded with 4600 CAD for 10 selected graduated students

Outstanding Project Award in Contest of Implementation

EECS, NTHU, 2018

- Top 10 of Research project competition with more than 250 student competitors

RELATED SKILLS

Programming: Python, SQL, Async Python, PyTorch, TensorFlow, Pytest, FastAPI, Flask, MATLAB, SystemC, $\text{\LaTeX}$

Cloud & Data: Azure, Azure Databricks, Apache Spark, PySpark, Apache Iceberg, Azure ML, MLflow, Data Pipelines

AI/ML: LangChain, AutoGen, CrewAI, Semantic Kernel, OpenAI Agents, RAG, FAISS, Embeddings, Vector Databases

Tools: gRPC, Docker, Kubernetes, Terraform, GitLab CI/CD, GitHub Actions, Jira, Linux

Backend: Microservices, REST APIs, Observability, Telemetry, Model Evaluation, Monitoring, Distributed Systems